

2 Broccoli is a vegetable plant that is grown as a crop in many parts of the world.

Broccoli is susceptible to a disease called clubroot.

Genetic engineering can be used to produce clubroot-resistant varieties of broccoli.

Genetic engineering involves inserting resistance genes into the cells of the broccoli plants.

The resistance genes were obtained from linseed plants.

Scientists can use DNA profiling to identify the broccoli plants that have been successfully genetically engineered.

(a) Samples of DNA were extracted from the plants and subjected to a polymerase chain reaction (PCR) before analysis.

(i) Explain why PCR was used on the DNA samples.

(2)

.....

.....

.....

.....

.....

.....

(ii) Taq (DNA) polymerase is used in PCR.

Describe the role of Taq (DNA) polymerase.

(2)

.....

.....

.....

.....

.....

.....

(b) After PCR, gel electrophoresis was used to separate the DNA fragments.

(i) Which type of enzyme is used to produce DNA fragments?

(1)

- ☐ **A** integrase
- ☐ **B** restriction enzyme
- ☐ **C** reverse transcriptase
- ☐ **D** RUBISCO

(ii) The results of the gel electrophoresis were used to identify the plants that contained the resistance genes.

Deduce the types of plants that would have supplied the DNA samples for this analysis.

Give reasons for your answer.

Use the information in the question to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 2 = 8 marks)